

A white Carolina Dory boat is docked at a wooden pier. The boat has a white cabin and a steering wheel. A person is standing on the pier behind the boat. In the background, there is a house with a balcony and some greenery. The sky is blue with some clouds.

Photo Essay – Building a **Key Largo**

Carolina Dory

Plans for this boat may be found at:

<http://spirainternational.com/>

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The Carolina Dory is a specialized type of craft derived from the original Grand Banks styles after the advent of power. It was quickly learned that the built-in rocker on the bottoms of traditional dories was fine for the very limited speeds possible with oar or sail power, but only held the boats back once gasoline and diesel engines became available. As a result, builders straightened out the bottoms to allow the boat to plane, and the Carolina Dory was born. These sturdy craft were originally used as commercial fishing boats in the Mid-Atlantic States, especially in the inland and near-shore waters around the Carolinas. Unlike modern power boats, Carolina Dories don't require lots of power to move along at quite a good clip. They're economical and exhibit their traditional dory heritage by performing quite well and safely in choppy conditions.

Larger Carolina type dory skiffs are popular and getting more so. The Key Largo is a classic Carolina Dory including the well mounted outboard. For economical operation in shallow water, there is no better boat. This dory was designed with the professional guide in mind able to handle the rough stuff but also ideal for sneaking into knee deep flats to fish. The Key Largo can handle four fishermen with ease and will operate in less than a foot of water. With a 25 to 40 HP motor it'll move out on a plane and with a 65, you'll fly. Boats like this also operate safely offshore some heading 75 miles out in the gulf to fish for grouper and snapper. A more economical and versatile 24 footer is hard to find.



Like all Spira International boats, the Key Largo is built using frames made of construction grade lumber. The frames use all straight elements so are a snap to build, even for an inexperienced carpenter. A simple beam type strong-back jig is used for this boat as well. Notice the



easily made lap joints where the frame elements are joined. These joints are glued with a modern waterproof, stronger-than-the-surrounding-material type adhesive, typically epoxy or a polyurethane adhesive.



Once the frames are all in-place and temporarily attached to the strongback, the keelson, running right down the center of the bottom, is attached tying the frames all together. This is a frame element made from dimensional lumber, that is, lumber available in

finished size, whether in inch or metric standard size.

In this photo, the chine logs are also in place. These are the boards tying the frames together where the bottom meets the sides. These are thinner than the rigid frame and keelson so that they may be more easily bent. The frames are notched using a back saw creating the form fitting seat for the chine logs. These are then glued and screwed in-place giving the plywood planking a good place to anchor when it is later attached.



At the gunwale edge of the frames, an additional longitudinal element is added called the sheer clamp. Like the chine log, the sheer clamp will serve as an anchor to attach the plywood covering to. It is likewise seated in notches the builder cuts in the frames,



then glued and screwed in-place. In this view you also see the stem forming the bow angle of the boat. This is also a piece of standard sized lumber attached to the keelson.



After fairing the frames, the framing itself is used as a pattern to cut out plywood planking to cover the framing. The ply is cut a little oversized, then glued and screwed in-place. Note that simple butt-joints are used where the ply panels come together. On the

inside, the joint is covered with a plywood butt-block to ensure the joint is strong and water-tight.

The side planking is followed by the bottom planking. Then the screw (or nail) heads, any drilled holes, and the plywood joint seams are filled using a simple wood filler that may



be made using glue and sawdust to ensure the outside surfaces are flush and smooth.



After filling, the edges of the boat are rounded and fiberglass tape is applied using epoxy to strengthen and seal the joints. This is relatively simple and pays off in the long haul by adding substantial durability to

your boat. Properly made, these hulls will last a lifetime.



Next, the hull is covered with fiberglass and epoxy. Two layers of 6 ounce per yard cloth are recommended. This seals the wood, adds considerable strength and durability to the hull. By using a simple



technique I show you in the Ply on Frame Boat Construction manual that is included with every set of boat plans, the usual sanding chore is minimized after the epoxy cures. Instead of a rough, textured surface, the surface will harden smooth and shiny.



Next, you can sand, prime and paint your hull. If you are planning on keeping it in the water in a slip, particularly in seawater, then a good quality marine paint is recommended. If you are going to mostly trailer the boat, then an exterior grade polyurethane enamel is a great

choice. You can buy it at any hardware or warehouse home supply store. This builder elected to also add a non-fouling boat bottom paint, as his boat will be stored in the water.



Now it's time to flip the hull over. You can invite a big crew over and offer a barbecue and adult beverages, or use a simple block and tackle to turn your hull right-side up to start working on the inside.

It's a good idea to add flotation between the frames, but many builders elect to use a simple slatted wood decking. You can see that this builder has raised and crowned the deck by adding additional lumber atop the frames, creating a self bailing deck for safer offshore performance. This is not detailed on the plans but is an option many builders prefer.

This builder also added an inwale timber, just inside the hull from the sheer clamp. Though not required, it increases the hull's ruggedness and gives the boat a nice, finished appearance.





Finally, a rail cap is installed covering the tops of the frames. This can be of plywood or of solid lumber that may be painted or stained and varnished for a yacht-like appearance.

Light cabins, center consoles, thwart seats or other features can be added to the Key Largo to suit the builder's needs.

